

PRIMARY SOURCES:

Perlman, Radia. "An Algorithm for Distributed Computation of a Spanning Tree in an Extended LAN." *Proceedings of the Ninth Symposium on Data Communications (SIGCOMM '85)*, ACM, 1985, pp. 44–53, doi:10.1145/319056.319004.

This is the original paper explaining STP, and helped me understand the protocol on a deeper level. I will use this paper to better understand and explain STP.

Perlman, Radia. *Interconnections: Bridges, Routers, Switches, and Internetworking Protocols*. 2nd ed., Addison-Wesley, 1999.

Perlman's textbook explains networking principles in accessible language. This includes STP, and other protocols she made. This was used to deepen my understanding of STP.

Perlman, Radia. *Network Security: Private Communication in a Public World*. Edited by Charlie Kaufman and Mike Speciner, 2nd ed., Prentice Hall PTR, 2002.

This book, coauthored with Kaufman and Speciner, reflects Perlman's other contributions beyond networking. This establishes her as active in the technological industry.

Perlman, Radia, and Donald E. Eastlake 3rd. "Introduction to TRILL." *Internet Engineering Task Force*, docslib.org. Accessed 29 June 2026.

This document introduces TRILL, STP's successor, demonstrating her as an active member of the networking community.

Eastlake 3rd, Donald, et al. "Transparent Interconnection of Lots of Links (TRILL): Adjacency." *RFC 7177*, Internet Engineering Task Force, May 2014, www.rfc-editor.org/rfc/rfc7177.txt.

This text was mainly used to establish Perlman as active in the computer science and networking communities.

Perlman, Radia, et al. "Alternatives for Multilevel Transparent Interconnection of Lots of Links (TRILL)." *RFC 8243*, Internet Engineering Task Force, Sept. 2017, datatracker.ietf.org/doc/rfc8243/.

This RFC shows Perlman is still involved in protocol development in the modern day.

Zhai, Hongjun, et al. "Transparent Interconnection of Lots of Links (TRILL): Pseudo-Nickname for Active-Active Access." *RFC 7781*, Internet Engineering Task Force, Feb. 2016, datatracker.ietf.org/doc/rfc7781/.

This shows how Perlman is still active in protocol development.

Dunbar, Linda, et al. "TRILL: Directory Assistance Problem and High-Level Design Proposal." *RFC 7067*, Internet Engineering Task Force, Nov. 2013, datatracker.ietf.org/doc/rfc7067/.

Zhang, Mingui, et al. "Transparent Interconnection of Lots of Links (TRILL): Fine-Grained Labeling." *RFC 7172*, Internet Engineering Task Force, May 2014, datatracker.ietf.org/doc/rfc7172/.

[Perlman's contributions to the details of TRILL show her attention to detail in network protocols.](#)

IEEE. *IEEE 802.1D-1990: IEEE Standard for Local and Metropolitan Area Networks—Media Access Control (MAC) Bridges*. IEEE, 1990.

[This is the official IEEE standard that formalized STP,](#)

IEEE. *IEEE 802.1w-2001: IEEE Standard for Local and Metropolitan Area Networks—Amendment 2: Rapid Reconfiguration of Spanning Tree*. IEEE, 2001.

[This standard enhanced STP with faster convergence times. This shows how Perlman's protocols are constantly being iterated on.](#)

Grabosky, Chris. Personal interview conducted by Aarav Mehta. 26 June 2026.

[This was a direct interview with a former networking professional, now employed at MongoDB in sales. This shows the lasting impacts of STP,](#)

SECONDARY SOURCES

Rosenheim, David. "An Unassuming Giant in Her Field: Radia Perlman's Work Speaks Volumes." *Internet Hall of Fame*, 8 Mar. 2021, internethalloffame.org/2021/03/08/unassuming-giant-her-field-radia-perlmans-work-speaks-volumes/.

[This Internet Hall of Fame profile provides biographical and technical context, positioning Perlman as a foundational figure of the internet.](#)

Kenney, Alex. "Radia Perlman Doesn't Want to Be Called 'Mother of the Internet.'" *The Atlantic*, Mar. 2014, www.theatlantic.com/technology/archive/2014/03/radia-perlman-dont-call-me-the-mother-of-the-internet/284146/.

[This interview explores Perlman's own perspective on her legacy, and her resistance to gendered narratives.](#)

Cryan, Rosemarie. "How Tech Pioneer Radia Perlman Overcame Bias to Invent a Core Component of the Internet." *Fast Company*, 19 Mar. 2020, www.fastcompany.com/90615239/radia-perlman-internet-pioneer-gender-bias.

[This article analyzes how Perlman navigated gender bias in the 1970s and 1980s in the computer science field.](#)

Pasik, Aaron. "Why IEEE Fellow Radia Perlman Hates Technology." *Computerworld*, 22 Apr. 2011, www.computerworld.com/article/2743693/why-ieee-fellow-radia-perlman-hates-technology.html

This interview shows Perlman's perspective on technology culture, and her design philosophy of elegant, user-centered design.

Busch-Vishniac, Ilene J., et al. "Radia Perlman." *Women in the National Inventors Hall of Fame*, edited by Jill S. Tietjen, Springer, 2024, pp. 1–15, doi:10.1007/978-3-031-75526-2_35.

This academic chapter places Perlman with other female inventors, providing context for her achievements and impacts.

National Inventors Hall of Fame. Inductee Profile: Radia Perlman. *National Inventors Hall of Fame*, invent.org. Accessed 29 June 2026.

This official National Inventors Hall of Fame profile documents Perlman's patents.

Internet Hall of Fame. Inductee Profile: Radia Perlman. *Internet Hall of Fame*, internethalloffame.org. Accessed 29 June 2026.

The Internet Hall of Fame's official page on Perlman establishes her as pivotal to the creation of the internet.

Lemelson MIT Program. "Radia Perlman." *MIT*, lemelson.mit.edu/resources/radia-perlman. Accessed 29 June 2026.

MIT's inventor profile connects Perlman to her educational background.

Tanenbaum, Andrew S. *Computer Networks*. 5th ed., Pearson, 2010.

Tanenbaum's standard networking textbook explains STP as a foundational concept.

Hafner, Katie, and Matthew Lyon. *Where Wizards Stay Up Late: The Origins of the Internet*. Simon and Schuster, 1996.

This narrative history of the ARPANET provides context for the early internet.

Abbate, Janet. *Recoding Gender: Women, Technology, and the Myth of Innovation*. MIT Press, 2012.

This article provides a theoretical framework for gender bias in computer science.

Hicks, Marie. *Programmed Inequality: How Britain Discarded Women Technologists and Lost Its Edge in Computing*. MIT Press, 2017.

While focused on Britain, Hick's analysis of how women were pushed out of computing gives context for Perlman's experiences with gender bias.

Margolis, Jane, and Allan Fisher. *Unlocking the Clubhouse: Women in Computing*. MIT Press, 2002.

[This study of gender in computer science directly addresses the challenges women faced in the field,](#)

Ensmenger, Nathan L. *The Computer Boys Take Over: Computers, Programmers, and the Politics of Technical Expertise*. MIT Press, 2010.

[This history of programming gives context to how computer science transitioned from female to male-dominated.](#)

NPR. "When Women Stopped Coding." *NPR: Planet Money*, 21 Oct. 2014, www.npr.org/sections/money/2014/10/21/357629765/when-women-stopped-coding.

[This audio essay/article traces the moment when female participation in computer science declined sharply.](#)

National Center for Women and Information Technology. *Women in Computing: Why Is the Percentage Dropping?* NCWIT, 2010.

[This analysis documents the underrepresentation of women in computing.](#)

Ashcraft, Catherine, and Joseph Breuer. *Women in Computing: Why Is the Percentage Dropping?* American Association of University Women, 2010.

[This study examines the barriers to women's participation in computing.](#)

Campbell-Kelly, Martin, et al. *Computer: A History of the Information Machine*. 3rd ed., Westview Press, 2013.

[This history of computing gives context for Perlman's work.](#)

IMAGES:

Defence Systems Agency. "ARPANET - MILNT Diagram 1984.jpg." *Wikimedia Commons*, Apr. 1984, commons.wikimedia.org/wiki/File:ARPANET_-_MILNT_Diagram_1984.jpg. Accessed 29 June 2026.

[This diagram of the ARPANET's topology shows the topology of the ARPANET in 1984, well into its creation.](#)

FastLizard4. "Interface Message Processor Front Panel.jpg." *Wikimedia Commons*, 29 Oct. 2011, commons.wikimedia.org/wiki/File:Interface_Message_Processor_Front_Panel.jpg. Accessed 29 June 2026.

[This photograph of an IMP control panel shows what ARPANET hardware looked like.](#)

UCLA and BBN. "ARPANET 1970 Map.png." *Wikimedia Commons*, 1970, commons.wikimedia.org/wiki/File:ARPANET_1970_Map.png. Accessed 29 June 2026.

[This diagram of the ARPANET's topology shows the topology of the ARPANET in 1970, one year after its creation.](#)

Jalisco Campus Party. "Radia Perlman (19995453218).jpg." *Wikimedia Commons*, 25 July 2015, commons.wikimedia.org/wiki/File:Radia_Perlman_(19995453218).jpg. Accessed 29 June 2026.

[This is a photo of Perlman at Campus Party 2015, sharing her work with others.](#)

---. "Radia Perlman (20175369862).jpg." *Wikimedia Commons*, 25 July 2015, commons.wikimedia.org/wiki/File:Radia_Perlman_(20175369862).jpg. Accessed 29 June 2026.

[This is another photo of Radia Perlman at Campus Party 2015.](#)

---. "Radia Perlman (20175426302).jpg." *Wikimedia Commons*, 25 July 2015, commons.wikimedia.org/wiki/File:Radia_Perlman_(20175426302).jpg. Accessed 29 June 2026.

[This is another photo of Perlman at Campus Party 2015.](#)

Scientist-100. "Radia Perlman (2009).jpg." *Wikimedia Commons*, 6 Aug. 2009, commons.wikimedia.org/wiki/File:Radia_Perlman_(2009).jpg. Accessed 29 June 2026.

[This is a headshot of Perlman from 2009.](#)

Karol508. "Broadcast Storm.png." *Wikimedia Commons*, 6 Dec. 2010, commons.wikimedia.org/wiki/File:Broadcast_storm.png. Accessed 29 June 2026.

[This is a diagram of a broadcast storm.](#)

ReneOFD. "Spanning Tree Topology.png." *Wikimedia Commons*, 23 Dec. 2005, commons.wikimedia.org/wiki/File:Spanning_tree_topology.png. Accessed 29 June 2026.

[This is a diagram of the STP protocol.](#)

“ShakataGaNai” Davis, John. "Network Switches.jpg." *Wikimedia Commons*, 24 July 2008, commons.wikimedia.org/wiki/File:Network_switches.jpg. Accessed 29 June 2026.

[This is a photograph of network switches](#)

BalticServers.com. "BalticServers Data Center.jpg." *Wikimedia Commons*, 17 Mar. 2013, commons.wikimedia.org/wiki/File:BalticServers_data_center.jpg. Accessed 29 June 2026.

[This is a photograph of a modern data center.](#)